

□ 6-DOF Robotic Arm — 2-Credit Project Summary (One Pager)

Project Title

Incremental CAD Design and Simulation of a 6-DOF Robotic Arm

Objective

Develop a **parametric CAD model of a 6-DOF robotic arm**, complete detailed design of **J1 (base)** as the critical subsystem, and **validate** kinematic feasibility in URDF/Gazebo. Optional **3D printing** will be used for structural and transmission prototypes. Design target: **2 kg payload, ~400 mm working radius**.

Background

- Software environment is mostly prepared (ROS2, MoveIt, Gazebo, devcontainers, scripts).
- Hardware, motors, and transmission ideas are defined; procurement and BOM planning mostly complete.
- No CAD assets exist yet; 3-DOF skeleton is not modeled.
- **Goal:** produce a structured CAD model first, focusing on one link/joint at a time to reduce workload while keeping parametric flexibility.
- Workload scoped for 2 credits, CAD-first, belt-only transmission, no springs.

Proposed Work

1. Parametric CAD Skeleton

- Build full 6-DOF skeleton, starting with **J1-J3 (base/shoulder/elbow)**.
- Define joint axes, link envelopes, motor mounts, and bearing placeholders.
- Keep all link lengths, offsets, and joint limits parametric for future iteration.
- Non-priority joints (J2-J6) initially modeled to conceptual/preliminary detail.

2. Detailed Subsystem Design

- **J1 (Base):** fully detailed for fabrication and assembly.
- Motor placement, belt sizing, wide-spaced bearings, thick ribbed plates for torsional stiffness.
- Other joints: envelopes/parametric placeholders; focus detailed design on one critical subsystem first.

3. 3D Printing & Prototyping (Optional-If Time)

- Print structural frames, pulley mounts, and belt components for validation.
- Ensure fit, assembly, and basic motion clearance; prototype loads lighter than 2 kg.

4. Simulation (if time)

- Export CAD to URDF/Gazebo (or ANSYS), starting with J1-J3.
- **Validate:**
 - Joint motion and workspace (~400 mm reach)
 - Axis directions, joint limits, self-collision detection
 - Placeholder J4-J6 for eventual full 6-DOF simulation.

Expected Deliverables

- Parametric CAD model of full 6-DOF arm.
- Detailed CAD of J1 (base) subsystem.
- Conceptual/preliminary CAD for non-priority joints (J2-J6).
- URDF/Gazebo model demonstrating J1-J3 motion.
- Short report documenting design rationale, iteration decisions, and simulation results.

Key Design Considerations

- Belt-only transmission, no springs.
- 3D-printed components for structure, mounts, and pulleys.
- Modular, parametric design for incremental build and iteration.
- Maintain 2 kg payload and ~400 mm reach for kinematic feasibility.
- Prioritize mechanical clarity over full system assembly.